

MixUp Navigator

The current valid rule, with proof.
In Bahasa Melayu and English, even offline.

A validity-aware assistant that answers Malaysian civil servants only from circulars still in force, cites the clause and page, and flags what was cancelled.

Team: MixUp Navigator

University: _____

Members: _____ | _____ | _____

NEVER CITE A CANCELLED
CIRCULAR AGAIN

One question. The current rule. With proof.

MixUp Navigator answers from circulars still in force, cites circular, clause and page, and says what was cancelled, instead of quoting it.

Validity engine

Knows which pekeliling is in force, amended or cancelled (berkuat kuasa / dipinda / dibatalkan).

Cited answers

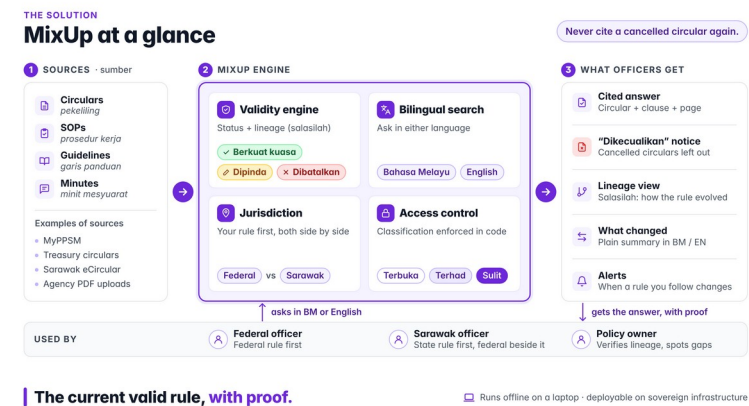
Every rule ends with [S1]: circular, clause and page you can open.

Right jurisdiction

Federal or Sarawak rule first, the other side by side.

Works offline

Runs on the officer's PC. Documents and questions never leave it.



Sources in: circulars, SOPs, guidelines and minutes from MyPPSM, Treasury circulars, Sarawak eCircular and agency files.

Out: a cited answer, the excluded cancelled circulars, lineage, what changed and alerts.

Finding the current rule is the real problem.

Circulars are amended and cancelled all the time, but the old file stays in folders, portals and chatbots.

1.3 million

federal civil servants follow these rules (ISEAS, 2026)

up to 445,000

officers already have Gemini via AI at Work 2.0 (Feb 2025): a general tool that does not know which circular is valid

395 agencies

and 56,000+ users store records in DDMS 2.0 (JDN via GovInsider): it stores files, it does not answer questions

Existing systems work in silos

DDMS 2.0
(storage)

X

MyPPSM
(HR circulars)

X

Treasury
circulars

X

Sarawak
eCircular

X

AI at Work
(general AI)

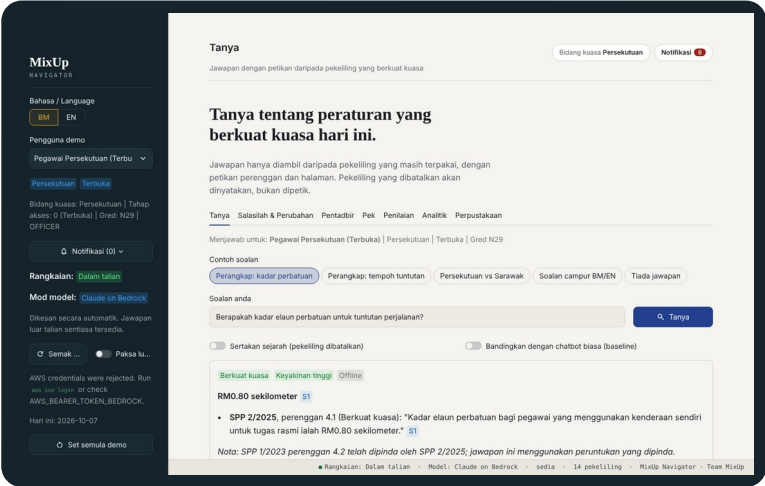
None of them answers: is this rule still valid, and what replaced it?

Why it's urgent

JPA is reviewing circulars older than 20 years and Act 867 (2025) requires reviews every 3 years: rules will change more often. The MyGov chatbot was shut down in Aug 2025 after giving nonsense answers: ungrounded AI fails in government.

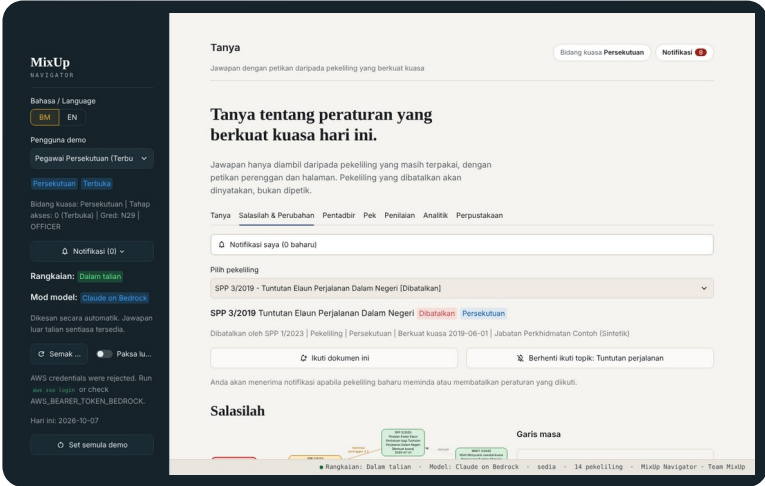
Ask. Check validity. Cite. Every screen is our working prototype.

One journey for every officer, in BM or English, fully offline.



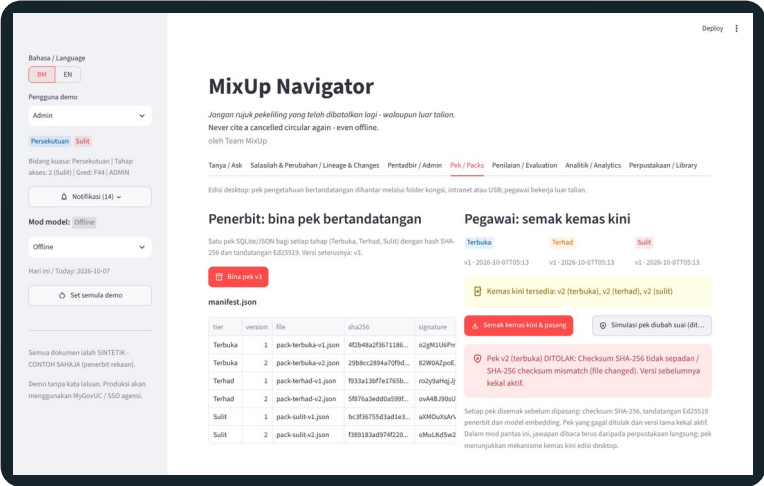
1. Ask and check validity

The mileage question is answered from SPP 2/2025 [S1]. The cancelled SPP 3/2019 is listed under Dikecualikan, never quoted.



2. Lineage and what changed

Salasilah 2019 -> 2023 -> 2025 shows who cancelled or amended what, with a plain BM/EN summary of the changes.



3. Signed updates

The Publisher ships signed packs. A tampered pack fails the SHA-256 check and is rejected; the previous version stays active.

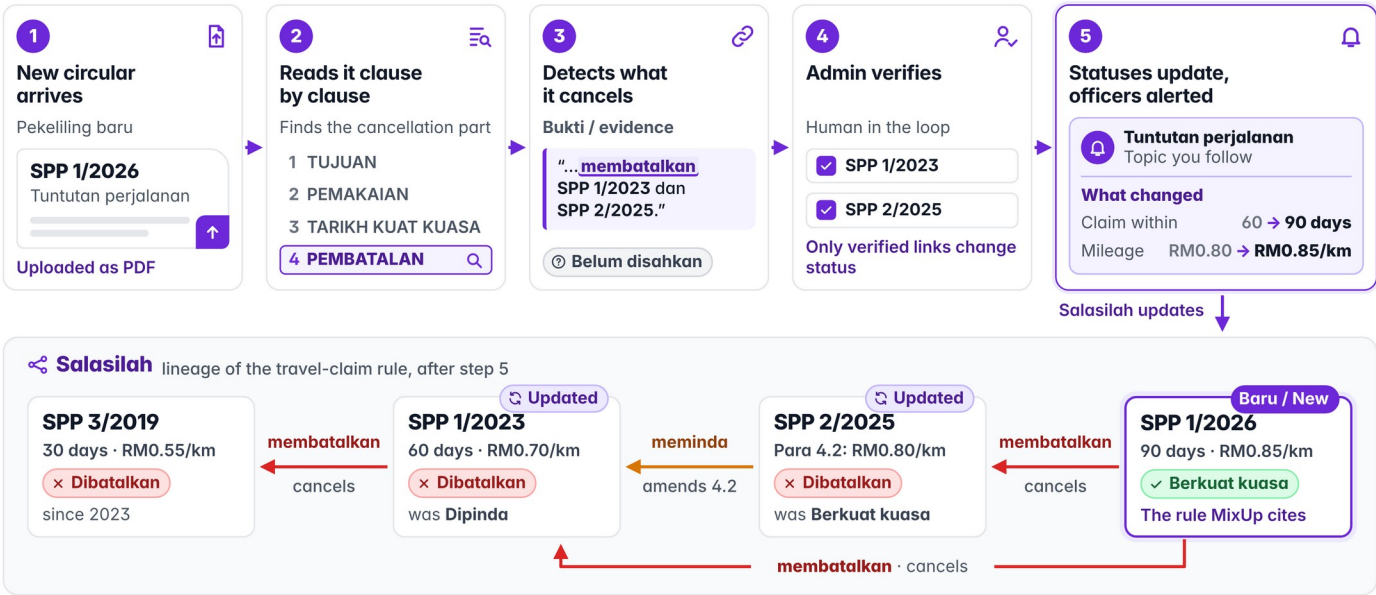
Officer app: ask, sources, lineage, updates. Publisher console: upload, verify relations, build packs. Validity engine underneath: statuses from verified relations only.

Chatting with documents is table stakes. Validity is the gap.

Gemini, NotebookLM, Copilot and Sarawak's ADAM AI already chat with documents. None, as far as public sources show, tracks which circular is still valid.

VALIDITY ENGINE

How MixUp keeps rules current



Rules change; MixUp keeps up – so officers don't have to.

Contoh sintetik / synthetic example

Always-latest guard

Answers only from in-force rules; cancelled ones are named, not quoted.

Federal vs Sarawak

Your jurisdiction first, the other side by side.

What-changed alerts

Clause-level diff and a plain BM/EN summary when a rule you follow changes.

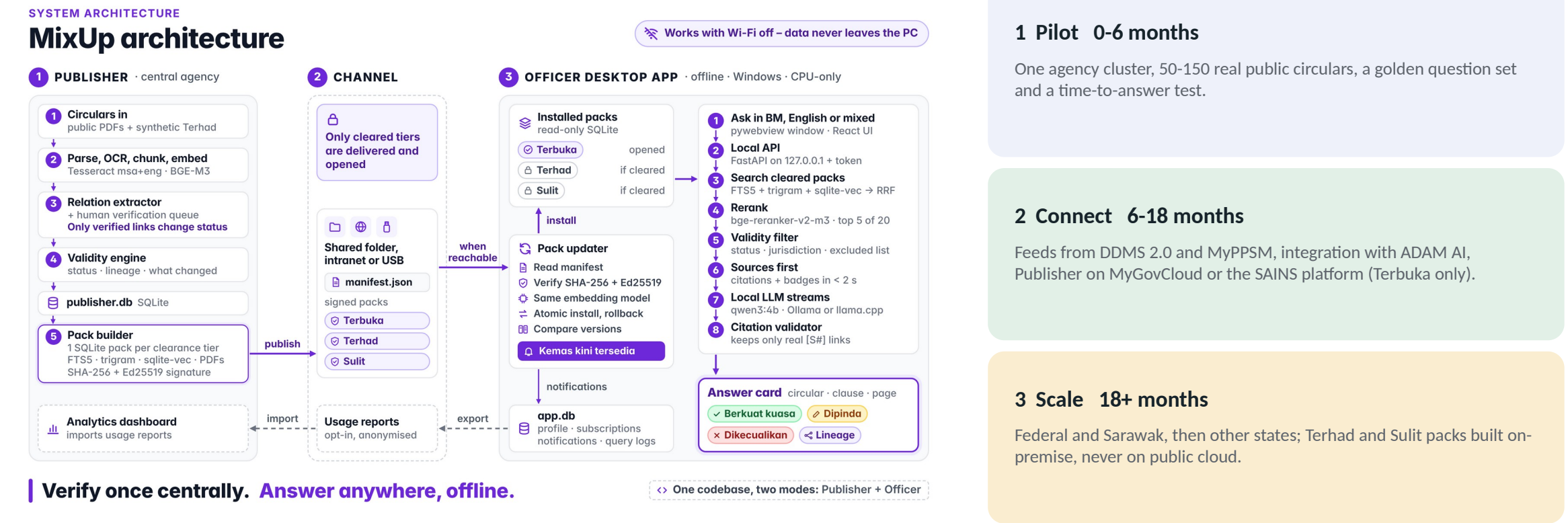
Offline, signed packs

Verified centrally, delivered signed, answered on the officer's PC.

91.7% -> 0% cancelled circulars cited as current: typical chatbot vs MixUp Navigator (30-question test, synthetic corpus)

Built in hours from parts that already exist.

Verify once centrally, deliver signed packs, answer anywhere offline. Public circulars can be curated in the AWS Malaysia Region; restricted tiers stay on-premise.



Correct, current answers for every officer.

Success means fewer wrong answers and rejected claims, not more chat.

ON OUR 30-QUESTION TEST (SYNTHETIC CORPUS)

91.7% -> **0%**

cancelled circulars cited as current

43% -> **83%** correct and current answers

Officers

The right rule in under a minute, with the clause to quote.

Agencies

Fewer rejected claims, reworks and complaints; audit trail of every answer.

Citizens

Consistent, correct answers from government, federal and Sarawak.

How we measure success

Cancelled-citation rate

Time to answer

Adoption per agency

Unanswered-question rate

Scale: 1.3 million federal civil servants, the Sarawak civil service, other states and GLCs. **Economic case:** minutes saved x questions per week x officers x hourly cost, with conservative inputs agreed with the pilot agency.

We diagnosed the real problem, built it, and measured it.

A working prototype, a fact-checked evidence base and a trust-first design.

01 Correct diagnosis

Our research found the gap is validity, not search. We verified every claim and dropped the weak statistics.

02 Built and evaluated

A working prototype with a 30-question evaluation against a plain chatbot, not slides alone.

03 Trust-first design

Human verification before any status changes, clearance enforced in code, offline by default.

Immediate next milestone

Pilot MixUp Navigator with one agency cluster on 50-150 real public circulars, and publish the before/after numbers.

Support we need

A pilot agency | access to a public circular corpus | feedback from JPA and the Sarawak civil service

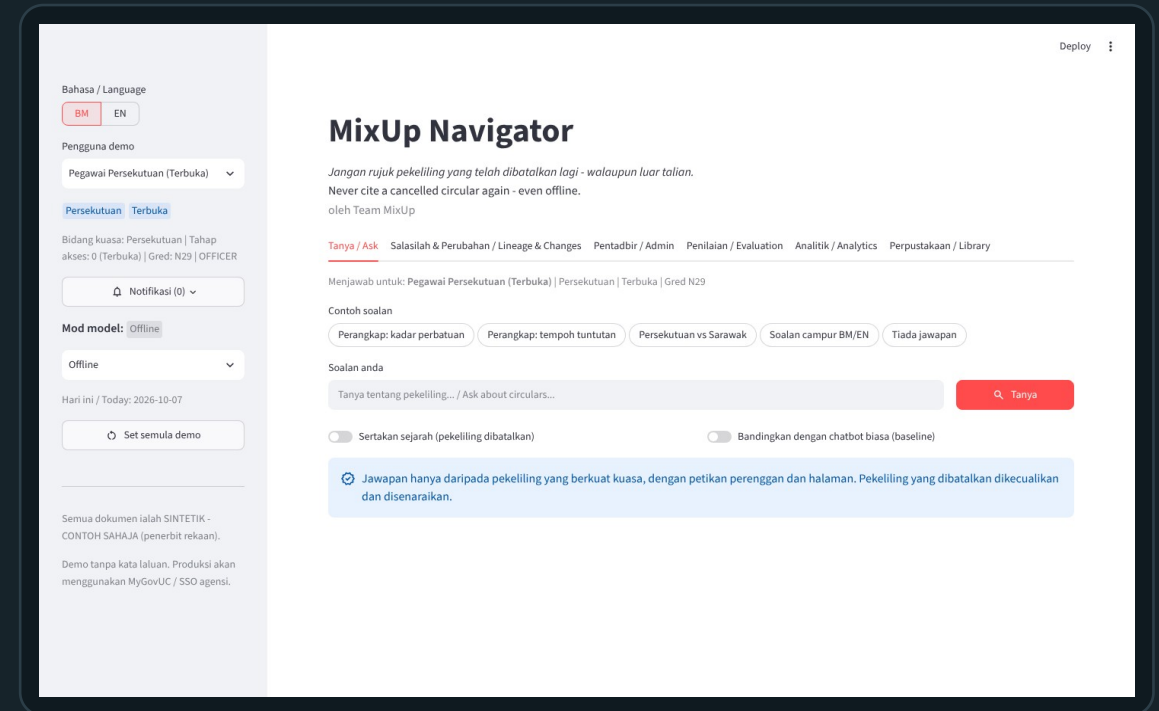
Every officer, the current rule, with proof. Never a cancelled circular again.

Thank you

MixUp Navigator prototype

- Officer mode: ask, sources, lineage, updates
- Publisher mode: upload, verify relations, build signed packs
- Evaluation page: typical chatbot vs MixUp Navigator
- Runs on a laptop, offline, with synthetic circulars

Repo: github.com/Rogerkkong/AWS-Kuching-Community-Day



What we checked, and where it comes from.

- Two teammate research reports (problem space; competitive analysis) plus an independent fact-check of every statistic; weak or unverifiable claims were dropped.
- ISEAS Perspective 2026/44: Malaysia's federal civil service of about 1.3 million.
- NST, FMT, Google Cloud (Feb 2025): AI at Work 2.0, up to 445,000 officers; pilot of 270 officers, 97% self-reported ~3.25 h/week saved.
- GovInsider / JDN: DDMS 2.0 used by 395 agencies and 56,000+ users.
- RTM, Berita Harian (2026): JPA review of circulars older than 20 years; Government Service Efficiency Commitment Act 2025 (Act 867) reviews every 3 years.
- Malay Mail, SoyaCincau (Aug 2025): MyGov beta AI chatbot shut down after nonsense answers.
- UKAS Sarawak (Jul 2026): ADAM AI for the Sarawak civil service; Borneo Post (Aug 2026): SAINS Sovereign AI Infrastructure Platform.
- Sarawak eCircular: Surat Pekeliling ICT Bil. 4 Tahun 2025 (AI guideline); AWS (2023): Malaysia Region, US\$6 billion by 2037.
- Evaluation: [prototype/eval/results/summary.md](#), 30 golden questions, synthetic corpus (SINTETIK - CONTOH SAHAJA).